

UNIFYING SPEECH AND LANGUAGE THROUGH CONTINUOUS DUAL-FUSION FOR MEDIUM-RESOURCE LANGUAGES: THE CASES OF HUNGARIAN AND GREEK

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ABSTRACT

In medium-resource languages like Hungarian and Greek, Automatic Speech Recognition (ASR) suffers from scarce paired acoustic data, whereas abundant text data can be leveraged via Large Language Model (LLM) priors. To compensate for this acoustic deficit, we propose a Dual-Fusion architecture that conditions ASR decoding on continuous acoustic information at both the prompt level and inside the LLM’s intermediate layers. Our model bridges a frozen acoustic encoder with pre-trained LLMs via both prompt-level embeddings and deep, continuous-space alignment across multiple intermediate decoder layers. To actively control the semantic-acoustic trade-off, our methodology introduces asymmetric downsampling to preserve fine phonetic boundaries, multi-layer gated cross-attention, attention-weighted layer-wise fusion of the acoustic encoder’s hidden states, and target text perturbation. Evaluated on diverse Hungarian and Greek datasets, our framework reduces Word Error Rate (WER) compared to both standalone acoustic decoders and acoustic-linguistic models like VoxKrikri, demonstrating consistent performance improvements across both languages.

Index Terms— Automatic Speech Recognition, Speech LLMs, Continuous Fusion, Medium-Resource ASR, Multimodal Alignment

1. INTRODUCTION

Automatic Speech Recognition (ASR) has advanced significantly with robust end-to-end models like Whisper [1] and data-efficient architectures like Canary [2], achieving state-of-the-art multilingual performance. However, these systems often degrade on medium-resource, morphologically complex languages such as Hungarian and Greek. We hypothesize this drop stems not solely from acoustic feature extraction, but partly from orthographic and morphological errors caused by a lack of deep, language-specific linguistic priors, consistent with evidence that external language models can improve low-resource ASR [3].

To bridge this gap, recent research couples pre-trained acoustic encoders with Large Language Models (LLMs) in Speech-Language Models (SLMs) [4], broadly following two integration strategies [5]: prompt-level approaches that prepend compressed acoustic embeddings as pseudo-text tokens, and cross-attention approaches that let text states attend to acoustic representations at intermediate layers. Early prompt-level strategies minimized architectural complexity: the SLAM-ASR framework [6] showed that a simple, trainable linear projector connecting a frozen encoder like Whisper to a frozen LLM yields competitive ASR. SALMONN [7] extended this with a learned Q-Former to bridge audio embeddings into the LLM space. We hypothesize that the rigid temporal pooling used by such prompt-only connectors risks destroying micro-phonetic boundaries required

for highly inflected languages; consistent with this concern, SLAM-ASR degrades under domain shift and acoustic perturbations [8].

For deeper semantic integration, cross-attention fusion was explored. The Whispering LLaMA framework [9] pioneered cross-modal generative error correction by integrating Whisper’s audio embeddings directly into the LLM via cross-attention adapters. A comparative connector study [10] reports that multi-head cross-attention, despite higher parameter cost than linear projectors, can yield better representational alignment than simple connectors. Furthermore, Whisper-Flamingo [11] established the efficacy of gated cross-attention to interleave frozen text blocks with continuous multimodal inputs, ensuring stable generation. In the speech domain, VoxKrikri [12] operates fully within a continuous latent space, fusing Whisper’s hidden decoder states with an LLM’s textual representations at a single selected layer, showing that intermediate layer injection yields robust cross-modal alignment for Greek.

A related question is which encoder representations to expose. Rather than passing only the final hidden state, SUPERB [13] established that different layers of a self-supervised speech encoder carry different phonetic and semantic information; follow-up work [14] shows learned weighted combination over all encoder layers can outperform single-layer extraction, and more recent work replaces static weights with input-dependent attention, e.g. Adaptive Layer Attention [15] and TASLA [16]. Finally, fused acoustic-LLM decoders often suffer the “lazy LLM” problem, ignoring acoustic information in favor of the text prior. We adopt a training-time token-substitution technique to mitigate this by weakening the model’s access to clean linguistic context during teacher forcing; this is related to, but distinct from, inference-time textual perturbation used to counter hallucination in multimodal LLMs [17].

Building on these findings, we propose a Dual-Fusion architecture that couples a frozen Whisper encoder with language-specialized LLM backbones for Hungarian and Greek, combining: (i) a *Dual-Fusion* strategy feeding acoustic information at both the prompt level and via cross-attention, with decoupled downsampling factors; (ii) *multiple injection layers* with *layer-wise fusion* of all Whisper encoder layers rather than only the final state; (iii) *gated cross-attention* scaling injected context with a learnable tanh gate; and (iv) *text perturbation*, substituting random tokens in the teacher-forced target sequence. We are not aware of a prior system that combines these within a single continuous, audio-conditioned architecture; we study whether doing so improves performance for both Hungarian and Greek, two typologically distinct medium-resource languages, and report a component-wise ablation (Section 4.5). Evaluated on multiple Hungarian and Greek corpora against standalone Whisper and VoxKrikri, our Dual-Fusion models yield consistent Word Error Rate reductions across both languages.

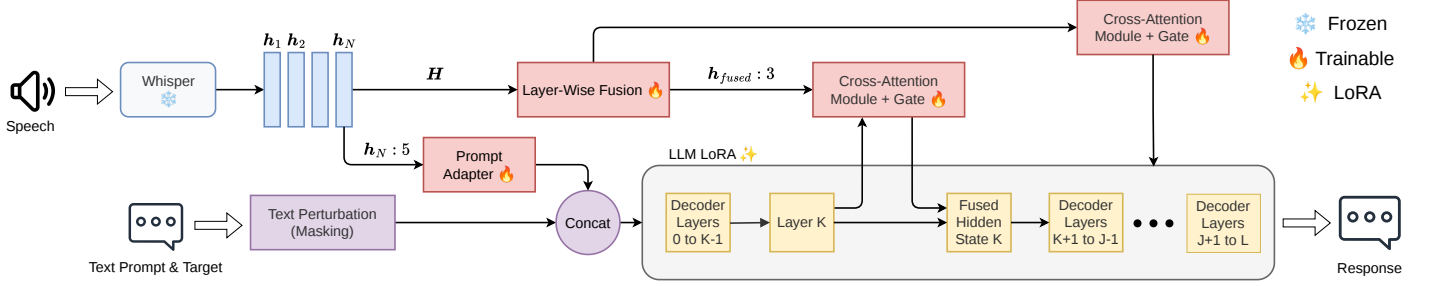


Fig. 1: Proposed Continuous Dual-Fusion ASR architecture: The speech instances are first processed by a frozen Whisper acoustic encoder, producing the acoustic hidden states $H = \{h_1 \dots h_N\}$. These states are processed in parallel by two distinct fusion pathways to bridge them into the LLM embedding space. In the prompt-level path, the final acoustic state (h_N) with dimensions $T \times D$ is downsampled by a factor of $k = 5$ and projected into the LLM space as prefix tokens of dimension $[T/5] \times D_{LLM}$ via an MLP adapter. Concurrently, in the cross-attention path, all encoder states H undergo Layer-Wise Fusion and Conv1D downsampling by a finer factor of $k = 3$ to form a phonetically rich acoustic representations of dimension $[T/3] \times D$. The LoRA-adapted LLM decoder follows; it processes the concatenated acoustic and text tokens, while aggregating the continuous acoustic representations via Gated Cross-Attention modules at multiple intermediate layers.

2. METHODOLOGY

The methodology couples a frozen Whisper encoder with a language-specific LLM (Racka-4B for Hungarian, Llama-Krikri-8B-Instruct for Greek) through prompt-level conditioning and intermediate cross-attention. An overview is given in Fig. 1.

2.1. Dual Fusion Strategy

To bridge the temporal mismatch between high-frequency acoustic frames and dense semantic tokens, we deploy asymmetric downsampling. In the *prompt path*, the final Whisper encoder state $h_N \in \mathbb{R}^{T \times D}$ is passed through a reshape operator to yield the downsampled state $\tilde{h}_N \in \mathbb{R}^{[T/5] \times 5D}$. A two-layer MLP then projects these downsampled representations into the LLM’s embedding space, yielding the continuous acoustic tokens $S_{prompt} \in \mathbb{R}^{[T/5] \times D_{LLM}}$. These are then concatenated into the LLM’s input sequence as prefix tokens:

$$\text{embed}(\text{prefix}) \parallel S_{prompt} \parallel \text{embed}(\text{text})$$

Crucially, this coarser downsampling factor ($k = 5$, 100ms resolution) prevents quadratic computational bloat in the LLM’s global self-attention sequence, while aligning with syllabic boundaries. Conversely, the *cross-attention path* injects speech-derived representations at selected layers using a finer Conv1D downsampler ($k = 3$, 60ms resolution). This preserves the micro-phonetic boundaries essential for highly inflected languages—specifically, rapid Hungarian agglutinative suffixes and unstressed Greek vowel transitions. Maintaining this denser resolution is computationally viable because it strictly localizes to cross-attention operations without expanding the LLM’s global sequence length. Ablations (Section 4.5) confirm $k = 3$ balances this model’s performance against the prohibitive cost of $k = 1$.

2.2. Multiple Injection Layers as Persistent Acoustic Memory

Rather than injecting at a single depth like VoxKrikri, we inject cross-attention simultaneously at three intermediate decoder blocks targeted at roughly 30%, 60%, and 90% relative depth (blocks [10, 20, 30] of 32 for KriKri, and [15, 25, 35] of 36 for Racka).

Repeated injection maintains a persistent acoustic memory, re-grounding generation instead of drifting toward pure language-model continuation.

2.3. Layer-Wise Fusion of Acoustic Representations

We utilize all encoder hidden states $\{h_0 h_1 \dots h_N\}$, since lower states preserve phonetic detail and higher states carry semantic abstraction. A *LayerWiseAttention* module mean-pools each state’s representation over the time dimension and scores it with a shared MLP, producing an *utterance-level* combination:

$$h_{fused}^{(j)} = \sum_{l=0}^N \alpha_l^{(j)} h_l, \quad \alpha_l^{(j)} = \text{softmax}_l \left(v^\top \tanh(W \bar{h}_l) \right)$$

2.4. Gated Cross-Attention

Injecting acoustic context at multiple layers risks letting acoustic signals overwhelm the LLM’s own generative flow, especially for purely syntactic tokens (punctuation, function words). Following the zero-initialized gating principle from Flamingo [18], each injection layer scales its cross-attention output with a single learnable scalar gate $g^{(j)}$:

$$\tilde{h}_{LLM} = h_{LLM} + \tanh(g^{(j)}) C_{audio}$$

2.5. Text Perturbation

To counter the “lazy LLM” problem, where the decoder over-relies on the teacher-forced text prior and ignores the injected acoustic information, we introduce a dynamic text perturbation mechanism. During training, target-history tokens are independently masked (replaced by the padding token) with a probability of $p = 10\%$. By stochastically masking the textual history during teacher forcing, the decoder is compelled to actively attend to the continuous acoustic representations to accurately predict the subsequent token. We set $p = 10\%$ per the ablation in Section 4.5.

3. EXPERIMENTAL SETUP

3.1. Backbones and Injection Configuration

Both language configurations adopt the frozen Whisper-large-v3 [1] encoder (637M parameters). For Greek, the LLM backbone is Llama-KriKri-8B-Instruct [19] (8.2B base parameters, hidden dimension $D_{LLM} = 4096$, 32 decoder blocks); for Hungarian, elte-nlp/Racka-4B [20] (4.0B base parameters, $D_{LLM} = 2560$, 36 decoder blocks). The trainable bridge network dynamically scales to match each LLM’s hidden dimension space. Specifically, the bridge consists of an MLP adapter (19M for Hungarian, 21.5M for Greek), a shared Conv1D downsampler (5M for both), cross-attention modules (59M for Hungarian, 132M for Greek), and a layer-fusion module (1M for both). This brings the total trainable parameters in Stage 1 to 84M for Hungarian and 160M for Greek, approximately 1.78% of the full model size in both configurations. In Stage 2, low-rank adaptation adds 168M LoRA parameters on KriKri-8B and 133M on Racka-4B, while the Stage 1 bridge remains frozen.

3.2. 2-Stage LoRA Curriculum and Training Hyperparameters

In Stage 1, the LLM and Whisper encoder are fully frozen and only the bridge network is optimized, for 45000 steps, with learning rate 10^{-4} , an effective batch size of 64 (per-device batch size 4 with 4 gradient-accumulation steps across 4 GPUs), bf16 mixed precision, and the AdamW [21] optimizer under a cosine scheduler with warmup. In Stage 2, QLoRA [22, 23] ($r = 64, \alpha = 128$) adapts the LLM for an additional 45k steps, targeting the attention projections and the gated-MLP up/down/gate projections; the bridge network trained in Stage 1 is frozen for the remainder of training so that LoRA adapts the LLM to the now-fixed acoustic interface rather than co-adapting with a still-moving bridge.

3.3. Corpora

We evaluate our architecture on diverse speech corpora spanning formal, read, and spontaneous registers for both languages. For Greek, the datasets include Common Voice [24], FLEURS [25], HParl [26], TEDx [27], Logotypographia [28]. For Hungarian, we utilize Common Voice [24], FLEURS [25], Speech Massive [29], VoxPopuli [30], and Yodas [31]. The Hungarian training mixture also incorporates “Internal-1” and “Internal-2”, comprising approx. 1,500 hours of conversational telephony and broadcast data. Each corpus is split into train/validation/test following a fixed priority: the creators’ own official splits, if provided; otherwise, a leave-subject-out (LSO) split by speaker/subject ID, ensuring no speaker overlaps train and test, using a 60/20/20 ratio. Because dataset sizes vary by orders of magnitude, training batches are constructed via interleaved sampling across corpora (with a temperature factor of $T = 1.2$).

3.4. Preprocessing, Prompting, and Inference

Transcripts are normalized by removing non-spoken artifacts, such as speaker-tags and transcriber annotations. The text is further simplified, by standardizing punctuation and verbalizing numbers. We actively retain punctuation; this ensures the model is trained to actively predict punctuation and syntactic boundaries directly from the acoustic signal and linguistic context. Utterances are conditioned on transcription style prompts: [PROMPT_CLEAN] (accurate, clean transcript), [PROMPT_VERBATIM] (word-for-word transcript, disfluencies retained), or [PROMPT_FORMAL] (strictly

formal/official transcript). Each corpus is assigned one style matching its own reference-transcription convention, applied consistently at training and inference. During training, Whisper’s input features are also regularized with SpecAugment [32]. Inference uses beam search (beam width of 4) with a repetition penalty of 1.15 and a 512-token maximum constraint.

4. RESULTS AND DISCUSSION

We evaluate using standard Word Error Rate (WER), Character Error Rate (CER), and Normalized Word Error Rate (N-WER), which is the error rate after stripping punctuation and casing variations (using the Whisper text normalization). Because standard WER penalizes missing punctuation, our strong results across both metrics indicate the architecture actively predicts syntactic boundaries from acoustic prosody. However, we use N-WER as our primary comparative metric to align with the VoxKrikri baseline.

4.1. Greek: Comparison Against Baselines

Table 1 compares against VoxKrikri (full-context, offline mode) and Whisper-v3 across five public Greek corpora. Our model outperforms VoxKrikri on Common Voice ($\Delta = -4.74$), HParl ($\Delta = -4.81$), and Logotypographia ($\Delta = -2.80$). FLEURS is the one corpus where VoxKrikri’s simpler single-injection design wins ($\Delta = +1.57$), which qualitative analysis suggests stems from FLEURS’s dense foreign named entities that our LLM’s linguistic prior occasionally mis-transliterates (Section 4.4).

Table 1: Greek ASR performance presented as WER / CER / N-WER (%). Δ_W compares our N-WER to the best competitive baseline

Dataset	Ours	Δ_W	Vox (full)	Whisper
Common Voice(19h) [24]	7.3 / 5.1 / 3.0	-4.74	12.04	13.98
FLEURS(12h) [25]	10.9 / 9.9 / 4.8	+1.57	9.3	11.02
HParl(120h) [26]	8.2 / 8.2 / 4.7	-4.81	13.01	16.99
Logotypographia(61h) [28]	6.6 / 6.6 / 2.0	-2.80	9.4	10.86
TEDx(29h) [27]	17.8 / 13.9 / 9.5	-	-	12.3

4.2. Hungarian: Comparison Against Baselines

Table 2 compares the model’s performance across all five public Hungarian corpora against Whisper large-v3. We deliberately exclude the Whisper-Hu [33] finetuned model due to disclosed test-set contamination on VoxPopuli, Common Voice, and Speech-Massive. Our model outperforms Whisper on four of the five comparable corpora, with the largest margins on Common Voice ($\Delta = -7.1$). The sole loss is Yodas, where plain Whisper remains stronger (11.3 vs. 6.2, $\Delta = +5.1$). This loss likely reflects reference provenance rather than a real shortfall: Yodas transcripts are pseudo-labeled by faster-whisper-large-v3 itself, so comparing us to zero-shot Whisper on Whisper-generated references is close to self-referential evaluation. Crucially, to isolate our architectural improvements from the proprietary data advantage, Table 3 details the performance of a model variant trained exclusively on the public corpora. Even without the 1,500-hour internal dataset, the public-only model suffers only a marginal degradation, maintaining a strict advantage over the Whisper baseline across the four comparable datasets and confirming the architectural efficacy of Dual-Fusion.

Table 2: Hungarian ASR performance presented as WER / n-WER / CER (%). Δ_W represents the N-WER difference between our model and the original Whisper baseline.

Dataset	Ours	Δ_W	Whisper
Common Voice(126h) [24]	7.4 / 6.0 / 2.0	-7.1	13.1
FLEURS(14h) [25]	10.4 / 7.7 / 2.9	-5.7	13.4
Speech Massive(6h) [29]	32.2 / 14.1 / 8.1	-5.3	19.4
VoxPopuli(60h) [30]	10.0 / 7.8 / 4.9	-6.7	14.5
Yodas(114h) [31]	14.0 / 11.3 / 7.1	+5.1	6.2

Table 3: Impact of proprietary data on n-WER (%): **Ours (Public)** represents the model trained exclusively on public corpora. Δ indicates the error rate degradation when the internal datasets have been removed

Model Configuration	CV	FL	SM	VP	YD
Ours (Public)	7.0	10.0	16.9	8.8	12.7
Δ vs. Ours (Full)	+1.0	+2.3	+2.8	+1.0	+1.4

Note: CV (Common Voice), FL (FLEURS), SM (Speech Massive), VP (VoxPopuli), YD (Yodas)

4.3. Cross-Attention Analysis

To empirically substantiate the depth specialization hypothesized in Section 2.2, we measured attention entropy ($\bar{H}$) and Pearson diagonality (r) over 1k random Common Voice samples across Racka’s three injection layers [15, 25, 35]. As visualized in Fig. 2, our quantitative analysis confirms distinct functional roles for each depth: (i) the earliest layer (15) exhibits high entropy ($\bar{H} = 0.951$) and moderate, highly variable diagonality ($r = 0.534$), for coarse salient event detection; (ii) the middle layer (25) achieves sharp temporal focus, characterized by the lowest entropy ($\bar{H} = 0.759$) and highest monotonic diagonality ($r = 0.887$). Notably, this precise grounding occurs at $\sim 69\%$ relative depth of Racka’s 36 decoder layers, corroborating independent findings for VoxKrikri [12]; and (iii) the latest layer (35) shows a relaxation in temporal tracking, with elevated entropy ($\bar{H}^{\text{norm}} = 0.788$) and reduced diagonality ($r = 0.704$), confirming its role in diffuse, non-monotonic global re-grounding.

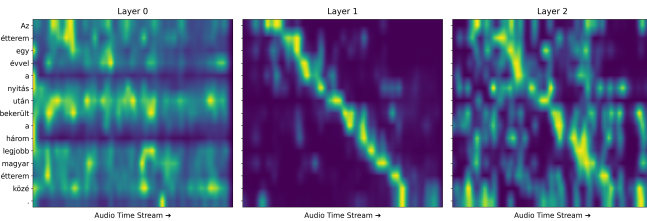


Fig. 2: Word-level cross-attention alignment for a representative Hungarian test utterance across the three injection layers. Temporal tracking sharpens strictly in the middle layer (Layer 25).

4.4. Limitations

Our qualitative analysis surfaces a recurring failure mode: because the fused LLM carries strong linguistic priors, it sometimes acts less like a verbatim transcriber and more like a “copy editor”. Observed behaviors include silently dropping hesitation markers

and dysfluencies, normalizing colloquial contractions to their formal written form, occasionally completing thoughts on abruptly cut-off audio, and mis-transliterating foreign proper names, inflating WER on disfluency-rich, verbatim-style corpora even when the output is fluent and largely faithful in meaning.

Finally, our evaluation is also limited to two medium-resource languages, a single frozen acoustic encoder (Whisper-large-v3); generalization to other languages, encoders, or truly low-resource settings remains untested.

4.5. Ablation Study

To isolate component contributions without prohibitive training costs, we evaluate ablations on the validation set at 12k steps (Table 4). As shown in Table 4, the Full Model achieves a baseline validation WER of 21.4%. Removing or altering any core component consistently degrades performance, validating our design: (i) removing the prefix audio tokens causes the largest loss (+3.3). It deprives the LLM’s early layers of global acoustic context. Most importantly, prefix tokens are visible to self-attention in every block from the first step; (ii) using only the final encoder layer degrades WER by 1.8, as intermediate layers retain phonetic detail [13]. Static, utterance-independent layer weights recover part of this gap but still degrades by +1.1, showing that adapting the mixture to each utterance brings a further gain; (iii) removing the zero-initialized tanh gate (+1.1) injects unstructured, randomly-initialized noise into the pre-trained LLM from the first update. The gate ensures stable, gradual integration of the acoustic signal; (iv) masking 0% of the target history increases WER by 1.1, as the model over-relies on text priors for subword completion and ignores the acoustic pathway. Conversely, excessive masking (20–30%) destroys the linguistic context required for clean inference; (v) $k = 5$ matches the prompt-path resolution and adds less temporal detail (+0.4). $k = 1$ gains only 0.2 while tripling the key/value length of every cross-attention module, so $k = 3$ offers the best accuracy–cost trade-off

Table 4: Ablation study for Hungarian on the validation set, presented as WER. Δ_W represents the WER degradation compared to the Full model.

Configuration	WER (%)	Δ_W vs. Full
Full model	21.4	–
Without Prompt Injection	24.7	+3.3
Without Layer-Wise Acoustic Fusion	23.2	+1.8
With Input-Independent Layer Weights	22.5	+1.1
Without Gated Cross-Attention	22.5	+1.1
Without Text Perturbation, $p = 0\%$	22.5	+1.1
With Text Perturbation, $p = 20\%$	22.1	+0.7
With Text Perturbation, $p = 30\%$	22.5	+1.1
Cross-Attn Downsampling Factor $k = 5$	21.8	+0.4
Cross-Attn Downsampling Factor $k = 1$	21.2	-0.2

5. CONCLUSION & FUTURE WORK

We presented a Continuous Dual-Fusion architecture coupling a frozen Whisper encoder with language-specialized LLMs for medium-resource ASR in Hungarian and Greek, using multi-layer gated cross-attention, layer-wise acoustic pooling, and text perturbation to ground generation in the acoustic signal without discarding

linguistic priors. Across five Hungarian and five Greek corpora, our model reduces WER over standalone Whisper on all but one Hungarian corpus (Table 2) and beats the strongest published baseline, VoxKrikri, on three of four comparable Greek corpora (Table 1).

We plan to: (i) incorporate acoustic data from higher-resource languages (e.g., English, German for Hungarian and English, Cypriot for Greek) to strengthen shared representations while confining foreign-language exposure to the encoder/cross-attention pathway and monitoring for code-switching leakage; (ii) use TTS-synthesized utterances to expose the linguistic side to target-language words absent from our corpora, particularly for morphologically rich Hungarian; and (iii) explore an audio-forecasting head predicting the next downsampled acoustic embedding from the LLM’s hidden states at injected audio-token positions, which could serve as transcript-free self-supervised pretraining over unlabeled speech, directly addressing the paired-data scarcity motivating this work.

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